Sergei Volkov

Researcher & Scientific Data Engineer · Hydrophysics, Limnology & Data Systems

Berlin, Germany · Tel: +49 15751115112 · Email: taranarmo@gmail.com
ORCID: 0000-0002-5385-1541 · GitHub: github.com/taranarmo · LinkedIn: linkedin.com/in/taranarmo
Work Authorization: EU Blue Card holder (>2 yrs) · Unrestricted German Work Permit (incl. Freiberufler / Commercial)

Researcher and Scientific Software Engineer bridging empirical physical science (physical limnology, convective turbulence, thermodynamics) and modern high-performance data engineering (lakehouse architectures, distributed data pipelines, Linux systems, and declarative NixOS package maintenance). Co-author of 5 peer-reviewed publications on convective turbulence and thermal stratification in boreal lakes. Extensive industry track record architecting enterprise-scale data infrastructure (Zalando, Playrix, AppIQ) alongside academic research (IGB Berlin, Northern Water Problems Institute).

PROFESSIONAL & RESEARCH EXPERIENCE

Freelance Data / MLOps Engineer | AppIQ
Jan 2026 – Present
Architecture & Implementation of ML-driven Experimentation Platform

- Architected and deployed a self-hosted Lakehouse (**DuckDB/DuckLake on AWS S3**) via **Terraform**, delivering sub-second analytical query latency while eliminating Spark operational overhead.
- Built low-latency ML model serving infrastructure and automated training/evaluation pipelines for real-time model inference.
- Designed high-throughput event ingestion and user tracking pipelines, enabling automated A/B test assignment and real-time metric calculation.
- Standardized declarative infrastructure-as-code and reproducible deployment workflows using **Terraform** and **Nix/NixOS**.

Sabbatical & Open Source Engineering | Self-Directed
Apr 2025 – Sep 2025
Open Source Contributions & Advanced Technical Upskilling

- Dedicated sabbatical for intensive open-source contributions to **NixOS** (`nixpkgs`), packaging and maintaining standalone **Apache Airflow**, **Julia**, and **Databricks CLI**.
- Conducted architectural research on embedded analytical engines (DuckDB) and prepared for advanced machine learning training.

Data Engineer | Zalando SE
Oct 2024 – Mar 2025
Central Data Platform & Analytics Engineering Services

- Developed and maintained business-critical ETL/ELT pipelines using **Managed Workflows for Apache Airflow (MWAA)** and **Databricks Delta Lake**, supporting 15+ internal stakeholder teams across logistics and finance.
- Engineered automated CI/CD deployment pipelines for internal Python helper packages and notebook workloads across staging and production Databricks environments.
- Enforced schema evolution standards and automated data quality validation gates, significantly reducing downstream pipeline failures and data incidents.

Data Scientist & Cloud Architect | Crocus Labs GmbH
Aug 2023 – Feb 2024
IoT Platform Architecture & Computational Photometry

- Architected scalable cloud infrastructure on **AWS (IoT Core, Lambda, S3, DynamoDB)** for real-time telemetry ingestion from smart greenhouse LED hardware.
- Developed a Python raytracing simulation to model 3D light distribution and optimize LED chip array layouts based on raw photometric emitter data.
- Engineered a containerized IoT test harness with virtualized sensor clients and an **MQTT broker in Docker**, enabling end-to-end integration testing without physical hardware.

Data Quality Engineer / Analytics Engineer | Playrix
Nov 2019 – Feb 2023
Promoted across 3 roles (Data Analyst -> QA Engineer -> Data Quality Engineer) for 50M+ DAU platforms

- Led data quality assurance during the migration of legacy Big Data pipelines to a modern lakehouse architecture across **Spark, Hive, Redshift, and Snowflake**.
- Built dynamic SQL metaprogramming validation frameworks that automated data anomaly detection across billions of daily in-game event records.
- Designed and optimized executive **Tableau** dashboards and underlying dimensional data marts tracking player retention, monetization, and A/B test experiments.
- Managed a QA sub-team, establishing automated regression test suites and data verification protocols for engineering releases.

Doctoral Researcher / Research Data Engineer | **Leibniz Institute of Freshwater Ecology (IGB Berlin)** Jun 2021 – Mar 2023
DFG Project PycnoTrap: Hydrophysics, Particle Dynamics & Computer Vision

- Investigated the sinking, degradation, and hydrodynamic trapping of organic biomass across stratified lake pycnoclines under supervisor Dr. Georgiy Kirillin.
- Designed custom underwater twin-action camera frames to record stereo-video for in-situ 3D particulate velocity tracking in turbulent water columns.
- Architected containerized computational environments (**Docker, CMake, MATLAB, OpenCV, X11 forwarding**) to modernize and execute a legacy 3D particle-tracking codebase.
- Developed automated computer vision pipelines in **Python** to extract 3D velocity vectors and calculate vertical particulate carbon flux.

Research Assistant | **Northern Water Problems Institute (NWPI, Russian Academy of Sciences)** Jan 2016 – Nov 2024
Hydrophysical Modeling, Telemetry Analysis & Winter Limnology

- Conducted multi-year observational and numerical research on under-ice convective mixing and boundary-layer heat transfer in boreal lakes.
- Analyzed year-round high-frequency thermistor chain and ADCP telemetry to establish the presence and dynamics of energetic internal waves beneath ice cover.
- Calculated the full Reynolds stress tensor of convective turbulence from paired acoustic Doppler profiler observations.
- Co-authored 5 peer-reviewed publications in international journals (**Environmental Fluid Mechanics**, **Inland Waters**, **Geosciences**).
- Managed full research cycles: observational campaign logistics, sensor calibration, data processing pipelines, and manuscript preparation.

Teaching Assistant | **Petrozavodsk State University** May 2016 – Jul 2020
Department of Energy Supplies & Thermal Engineering

- Delivered lectures, computational seminars, and laboratory coursework in **Thermodynamics**, **Heat and Mass Transfer**, and **Applied Fluid Dynamics**.
- Mentored engineering undergraduate and graduate students in numerical modeling and experimental data analysis.

PEER-REVIEWED JOURNAL PUBLICATIONS

2021 **Full Reynolds Stress Tensor of Convective Turbulence Estimated with Paired Acoustic Doppler Current Profilers**
S. Bogdanov, G. Kirillin, S. Volkov, G. Zdorovennova
Earth and Space Science Open Archive (ESSOAr) · DOI: 10.1002/essoar.10507975.1

2019 **Fine scale structure of convective mixed layer in ice-covered lake**
S. Volkov, S. Bogdanov, R. Zdorovennov, G. Zdorovennova, A. Terzhevik, N. Palshin, D. Bouffard, G. Kirillin
Environmental Fluid Mechanics, 19(5), 1151–1165 · DOI: 10.1007/s10652-018-9652-2

2019 **Under-ice convection dynamics in a boreal lake**
D. Bouffard, G. Zdorovennova, S. Bogdanov, T. Efremova, S. Lavanchy, N. Palshin, A. Terzhevik, L. Vinnå, S. Volkov, A. Wüest, R. Zdorovennov, H. Ulloa
Inland Waters, 9(2), 142–161 · DOI: 10.1080/20442041.2018.1533356

2019 **Structure and dynamics of convective mixing in Lake Onego under ice-covered conditions**
S. Bogdanov, G. Zdorovennova, S. Volkov, R. Zdorovennov, N. Palshin, T. Efremova, A. Terzhevik, D. Bouffard
Inland Waters, 9(2), 162–178 · DOI: 10.1080/20442041.2018.1551655

2018 **Albedo of a Small Ice-Covered Boreal Lake: Daily, Meso-Scale and Interannual Variability on the Background of Regional Climate**
G. Zdorovennova, N. Palshin, T. Efremova, R. Zdorovennov, G. Gavrilenko, S. Volkov, S. Bogdanov, A. Terzhevik
Geosciences, 8(6), 206 · DOI: 10.3390/geosciences8060206

INTERDISCIPLINARY RESEARCH PROJECTS

Lake Pycnoclines Trap Organic Particles Forming Hot Spots of Accelerated Carbon Cycling (PycnoTrap) 2021 – 2023
Leibniz Institute of Freshwater Ecology and Inland Fisheries (IGB Berlin)

DFG-funded joint study examining biomass sinking, degradation, and hydrodynamic trapping across stratified lake columns. Built stereo-video computer vision tracking pipelines to derive 3D velocity vectors and calculate downward particulate carbon flux.

Lake Onego: Life Under the Ice 2015 – 2017
Northern Water Problems Institute (NWPI) / EPFL Limnology Center · Project Link

International collaborative field campaign investigating convective mixing regimes, solar radiation penetration, and biological dynamics beneath winter lake ice.

EDUCATION & ACADEMIC THESES

Specialist Degree in Energy Supplies (M.Sc. Equivalent) 2010 – 2015
Petrozavodsk State University
Diploma Thesis: *"Noniterative Heat Exchanger Calculation"* (Supervisor: Prof. Sergei Bogdanov)
Analytical calculation procedure for heat exchanger sizing, avoiding non-convergent numerical optimization algorithms.

Data Science & Machine Learning Bootcamp Oct 2025 – Dec 2025
Ironhack

Machine Learning Specialization 2017
Stanford University on Coursera

TECHNICAL, SCIENTIFIC & SYSTEMS COMPETENCIES

Domain Physics & Limnology:	Hydrophysics, Physical Limnology, Convective Turbulence, Stratified Flows, Heat & Mass Transfer
Field & Instrumentation:	Acoustic Doppler Current Profilers (ADCP), Thermistor Chains, CTD Profilers, Stereo-Cameras
Data Engineering & Platforms:	Spark, Databricks, DuckDB/DuckLake, Apache Airflow, Snowflake, dbt, Delta Lake, Redshift
Scientific Computing & ML:	Python (NumPy, SciPy, Pandas), MATLAB, Julia, OpenCV, Computer Vision, PyTorch, Scikit-Learn
Systems, DevOps & Infra:	Linux Administration, Docker, Terraform, Nix/NixOS, AWS, CI/CD (GitHub Actions, GitLab CI)
Open Source & Community:	Official Package Maintainer for NixOS (`nixpkgs`: Apache Airflow, Julia, Databricks CLI, plus 6 Python libraries, system utilities) · OpenStreetMap Contributor
Languages:	English (C1/C2 Full Professional), German (B2 Professional Working), Russian (Native)